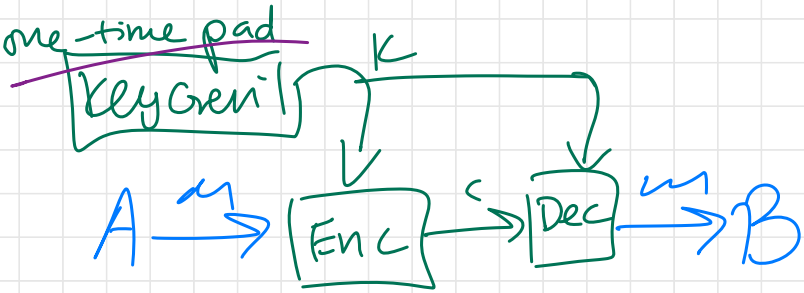
Encryption



Security

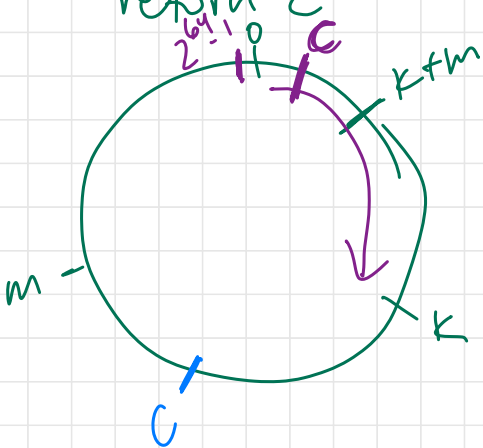
Whatever is ~~knowable~~ efficiently computable about m given c is also ~~knowable~~ efficiently computable without c.

$\rightarrow$ the distribution over m given c is uniform

ex $\underline{\text{Enc}(k, m)}$:

$c = (k + m) \bmod 2^{64}$

return c



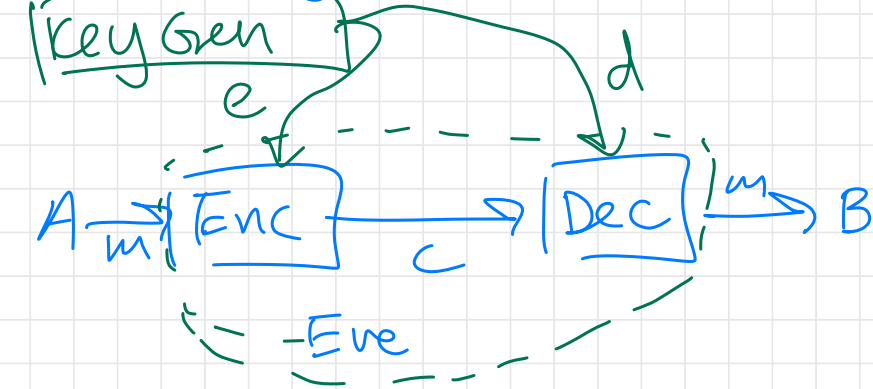
number line mod 2^{64}

ex $\mathrm{Enc}(k, m)$

$$c = (k \cdot m) \bmod 2^{64}$$

does not satisfy security prop.

1949: Security prop. only possible with perfectly random, one-time-pad k

notice: $|k| = |m|$

RSA Encryption